Kubernetes: Storage

VISHWANATH M S
VISHWACLOUDLAB.IN

Agenda

- **Storage**
- **Volumes**
- **Persistent Volumes (PV)**
- **Persistent Volume Claims (PVC)**
- **Examples with YAML**

Introduction to Kubernetes Storage

- Kubernetes provides abstractions to manage storage resources for containers.
- Containers are **ephemeral**, but apps often require **persistent storage**.
- Storage options:
  - EmptyDir
  - HostPath
  - NFS
  - Cloud-based volumes (EBS, Azure Disk, GCE Persistent Disk)

Describe Storage

- **Volume**: Directory accessible to containers in a Pod. Lifecycle is tied to the Pod.
- **Persistent Volume (PV)**:
  - Cluster-wide resource, provisioned by admin or dynamically.
  - Exists independently of Pods.
- **Persistent Volume Claim (PVC):**
  - A request for storage by a user.
  - Binds to a suitable PV.
- **StorageClass:**
  - Describes "classes" of storage (e.g., fast SSD, standard HDD).
  - Supports dynamic provisioning of volumes.

Lab: Volumes

Objective: Understand and use ephemeral volumes like emptyDir and hostPath

Lab Steps:

1. Create a Pod with emptyDir volume:

```
volumes:
  - name: cache-volume
    emptyDir: {}
```

2. Mount the volume inside a container at /cache

```
volumeMounts:
  - name: shared-data
    mountPath: /usr/share/nginx/html
```

3. Observe data persistence within Pod lifecycle (deleted once Pod is deleted)
4. Use hostPath to mount a specific directory from node

Why Not Just Volumes?

- **Volumes are tied to pod lifecycle → data lost if pod dies**
- **Not suitable for stateful apps (DBs, logs)**
- **Solution: Use Persistent Volumes (PV) and Persistent Volume Claims (PVC)**

What is a Persistent Volume?

- **A PersistentVolume (PV) is a piece of storage provisioned by an admin or dynamically created.**
- **It’s a cluster resource, independent of pods.**
- **Types:**
  - **NFS**
  - **iSCSI**
  - **AWS EBS**
  - **Azure Disk**
  - **HostPath (for local use)**

What is a Persistent Volume? - Example

```yaml
apiVersion: v1
kind: PersistentVolume
metadata:
  name: my-pv
spec:
  capacity:
    storage: 1Gi
  accessModes:
    - ReadWriteOnce
  hostPath:
    path: /mnt/data
```

Pod Using PVC

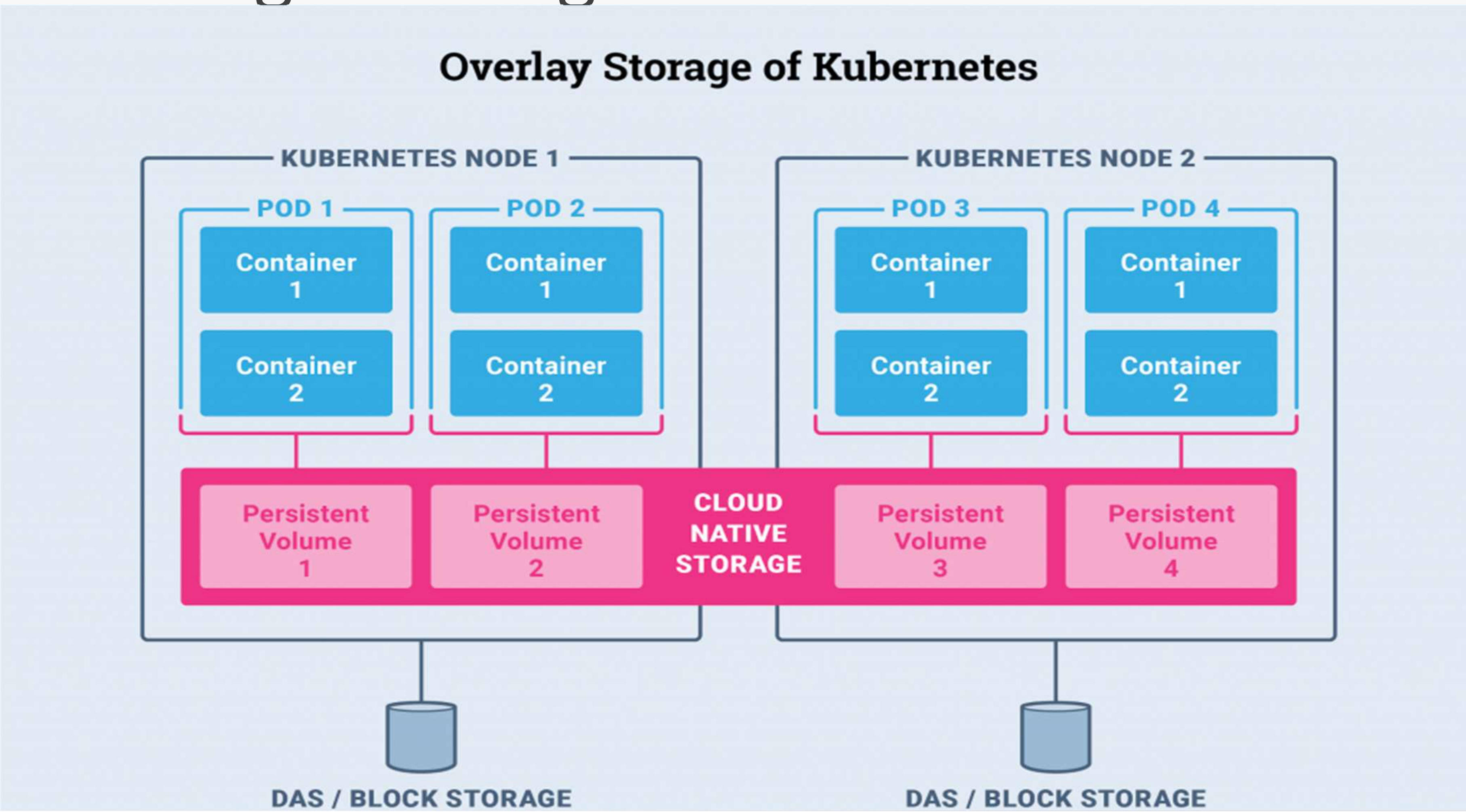
```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pv-pod
spec:
  containers:
    - name: nginx
      image: nginx
      volumeMounts:
        - mountPath: "/usr/share/nginx/html"
          name: my-storage
```

```
  volumes:
    - name: my-storage
      persistentVolumeClaim:
        claimName: my-pvc
```

Storage Binding Flow

- **Admin creates a PV (or uses dynamic provisioner).**
- **Developer creates a PVC.**
- **PVC binds to a matching PV.**
- **Pod uses PVC via volumeMount.**

Storage Binding Flow



LAB.IN

Types of Access Modes

Mode	Description
ReadWriteOnce	One node can read/write at a time
ReadOnlyMany	Many nodes can read
ReadWriteMany	Many nodes can read/write (e.g., NFS)